SESSION - XVI

Regression Model

Regression line or curve — Dependence of Y on X
Conversely . . . . . X on Y.

Regression Curve — Best approximates the dependence on the basis of sample data.

Given joint density $f(x,y)$ of rvs X and Y.

Design function $d(x) = E[Y|x]$.

* Since conditional distributions in practice are difficult to obtain, we restrict choices to specific class of functions

$$D = Y - d(x)$$

$$\text{Minimize } E(D^2) = E[(Y - d(x))^2]. \quad (1)$$

Linear Model of dependence

$$d(x) = a + bx$$

$$G(a,b) = E(D^2) = E[(Y - (a+bx))^2]$$

or $$G(a,b) = \sigma_Y^2 + b^2\sigma_X^2 + (\mu_Y - a - b\mu_X)^2 - 2b\rho\sigma_X\sigma_Y \quad (2)$$

To Minimize $G(a,b)$:

$$\left.\begin{aligned} \frac{\partial G}{\partial a} = 0 &\Rightarrow \mu_Y - a - b\mu_X = 0 \\ \frac{\partial G}{\partial b} = 0 &\Rightarrow 2b\sigma_X^2 - 2\mu_X(\mu_Y - a - b\mu_X) - 2\rho\sigma_X\sigma_Y = 0 \end{aligned}\right\} \quad (3)$$

Optimal Values of a and b

$$\hat{b} = \rho\frac{\sigma_Y}{\sigma_X} \text{ and } \hat{a} = \mu_Y - b\mu_X = \mu_Y - \rho\frac{\sigma_Y}{\sigma_X}\mu_X \quad (4)$$

Regression line of Y on X

$$\frac{y-\mu_Y}{\sigma_Y} = \rho \frac{x-\mu_X}{\sigma_X} \qquad (5)$$

Similarly regression line of X on Y

$$\frac{x-\mu_X}{\sigma_X} = \rho \frac{y-\mu_Y}{\sigma_Y} \qquad (6)$$

Square of Minimum Prediction Error

$$E(D^2) = G(\hat{a}, \hat{b}) = (1-\rho^2)\,\sigma_Y^2 \qquad (7)$$

$\rho^2 \neq 1$, then

$$\sigma_Y^2 = \rho^2 \sigma_Y^2 + E(D^2) \qquad (8)$$

Variance of Y attributable to linear dependence

Residual error that cannot be explained by linear dependence.

Note: The linear predictor is attractive as it requires knowledge of first two moments only.

Least Squares Curve Fitting

Given n points $\{(x_i, y_i) \mid i = 1, \cdots n\}$

Choose a and b so that

$$S = \sum_{i=1}^{n} [y_i - (a + bx_i)]^2 \text{ is minimum.} \qquad (9)$$

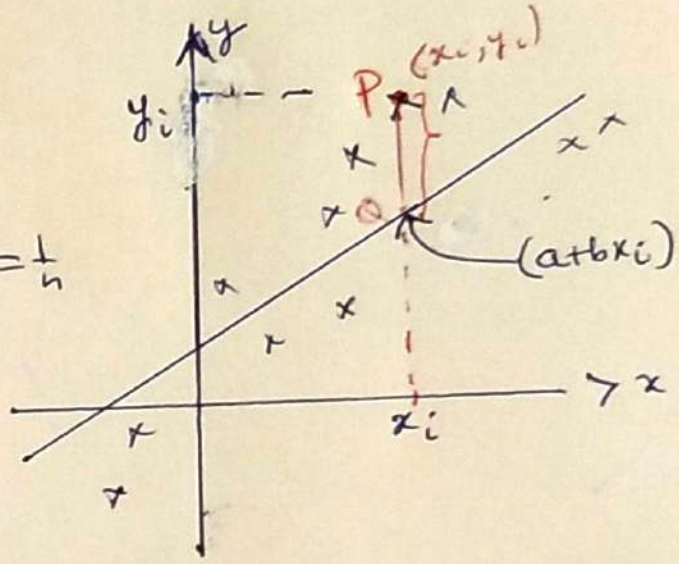
We get $a = \bar{y} - b\bar{x}$, $b = \dfrac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2}$ (10)

Normal Equations:

$$\frac{\partial S}{\partial a} = 0 \Rightarrow \quad n a + b \sum x_i = \sum y_i \qquad (11)$$

$$\frac{\partial S}{\partial b} = 0 \Rightarrow \quad a \sum x_i + b \sum x_i^2 = \sum x_i y_i$$

Probabilistic Counterpart



Two rv's (X, Y)

Joint pmf $P(X = x_i, Y = y_i) = \frac{1}{n}$

$i = 1, \dots n$

Regression line Y on X

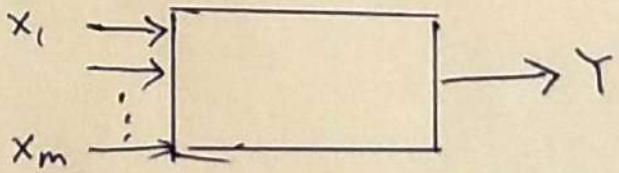
$$\text{Min } E(D^2) = \text{Min } E\left[\left(Y - (a + bX)\right)^2\right] \qquad (12)$$

$$= \text{Min} \sum_{i=1}^{n} \frac{1}{n} \left[y_i - (a + bx_i) \right]^2$$

More General Case

Input - output behaviour of stochastic system with m inputs

Independent rvs $\underset{\sim}{X} = (X_1, X_2, \dots, X_n)$



Appropriate prediction $d(\underset{\sim}{X}, \underset{\sim}{a})$

vector of parameters

parameterized family of functions

$$\frac{\partial}{\partial a_i} E\left[\left(Y - d(\underset{\sim}{X}, \underset{\sim}{a})\right)^2\right] = 0 \quad \text{for all } i \qquad (13)$$

Summary

Simple linear regression

$$E[Y \mid X = x] = a + bx \quad \text{for all } x \qquad (14)$$

R Vs Y_i, $i = 1, 2, \ldots n$ are of the form

$$\underline{Y_i = a + bx_i + \epsilon_i}, \quad i = 1, 2, \ldots n \qquad (15)$$

Random variables $\epsilon_1, \epsilon_2, \ldots \epsilon_n$ are errors which cause deviations about the linear relationship $a + bx$, $i = 1, \ldots n$.

The errors are independent and assumed to be normally distributed

$$E[\epsilon_i] = 0 \qquad var[\epsilon_i] = \sigma_i^2 \qquad i = 1, \ldots n$$

$\uparrow$ Recall CLT. (16)

It follows:

$$E[Y_i] = E[Y \mid X = x_i] = E[a + bx_i + \epsilon_i]$$
$$= a + bx_i = \mu_i \qquad (17)$$

$$V[Y_i] = V[Y \mid X = x] = V[a + bx_i + \epsilon_i]$$
$$= Var[\epsilon_i] = \sigma^2 \qquad (18)$$

$Y_1, \ldots Y_n$ are mutually independent because of mutual independence of $\epsilon_1, \epsilon_2 \ldots \epsilon_n$.

(From A. O Allen)

The Coefficients of Determination

Least squares fit to data

$$y = a + bx \qquad (14)$$

Consider goodness of fit :- For a point x_i, the value predicted $\hat{y}_i$ is given by

$$\hat{y}_i = a + bx_i \qquad (15)$$

* [Difference between observed and predicted value: $|y_i - \hat{y}_i|$ should be low for a good fit

Note $|y_i - \bar{y}| = |(y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})|$

Squaring, we get

$$\sum_i (y_i - \bar{y})^2 = \sum_i (y_i - \hat{y}_i)^2 + \sum_i (y_i - \bar{y})^2 + 2 \underbrace{\sum_i (y_i - \hat{y}_i)(\hat{y}_i - \bar{y})}_{=0}$$

$$\sum (y_i - a - bx_i)(a + bx_i - \bar{y})$$
$$= a \sum (y_i - a - bx_i) + b \sum x_i (y_i - a - bx_i) \qquad (16)$$
$$- \bar{y} \sum (y_i - a - bx_i) = 0 \qquad \text{(Normal eqns)}$$

Therefore

$$\underbrace{\sum (y_i - \bar{y})^2}_{\text{Total Variation}} = \underbrace{\sum (y_i - \hat{y}_i)^2}_{\mathrm{I}} + \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\mathrm{II}} \qquad (17)$$

$(y_i - \bar{y})$: [Deviation of observed value of Y from its sample mean

RHS

I Unexplained Variation

II Explained Variation

} by the least-squares line.

Note $(\hat{y}_i - \bar{y})$ Difference bet predicted value and sample mean

$$\hat{y}_i - \bar{y} = b(x_i - \bar{x}) \leftarrow \text{explained by least squares line}$$

II. Therefore $\sum_i (\hat{y}_i - \bar{y})^2$ is called Explained Variation.

I. $\sum (y_i - \hat{y}_i)^2$ is the residual sum of squares Unexplained Variation

(i) Should be as small as possible.
(ii) Zero if all observations lie on the fitted line.

We find

$$\text{Total Variation} = \text{Unexplained Variation} + \text{Explained Variation} \quad (19)$$

$$\text{Coefficient of determination} = \frac{\text{Explained Variation}}{\text{Total Variation}}$$

$$= \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} \quad (20)$$

lies bet 0 and 1

Better fit as it becomes closer to 1

Example (from Trivedi):
The failure rate $h(t)$ of a system is thought to be a power function of its age so that

$$h(t) = c\,t^d \qquad t \geq 0$$

A large no. of systems put on test and we divide ~~each~~ the number of failures in each

hourly interval by the number of surviving components to obtain an observed value of h_t as given below (see Table)

Soln.

$$h(t) = c\,t^{d} \qquad (21)$$

or $\ln h(t) = \ln c + d \ln t$

Writing $Y_t = \ln h_t$, $x_t = \log t$, we find (22)

$\sum y_i = 13.1343 \quad \sum x_i = 6.5598$

$\sum y_i^2 = 20.86416 \quad \sum x_i^2 = 5.2152$

$\sum x_i y_i = 10.42965$

From (22), the linear line of regression is

$$\boxed{y = a + bx} \qquad (23)$$

Questionable!

where $a = \ln c$ and $b = d$

The data provides

$\hat{a} = \log c = 0.008930$

or $\boxed{\hat{c} = 1.0208}$

and $\hat{b} = d = 1.9886$

Also $\sum (\hat{y}_i - \bar{y})^2 = 3.607 \qquad (24)$

$\sum (y_i - \bar{y})^2 = 3.613$

Coefficient of determination $= \dfrac{3.607}{3.613}$

$= 0.9983$

Hence the coeff of det is near perfect.

t, h	$h_t \cdot 10^3$	$Y_t = \log h_t$	$x_t = \log t$	$\hat{y}_i$
0.5-1.5	1.05	0.02119	0.000	0.008946
⋮	⋮	⋮	⋮	⋮
9.5-10.5	97.50	1.98900	1.000	1.997546
		13.1343	6.5598	

Multiple Linear Regression (From Allen)

The regression curve of Y on $\underset{\sim}{X} = (X_1, \cdots X_k)$ is of the form

$$E[Y \mid X = (x_1, x_2, \cdots, x_k)] = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k + \epsilon \quad (25)$$

for all $(x_1, \cdots x_k)$, where $\beta_0, \beta_1, \cdots \beta_k$ are constants

This model is called multiple regression model with k explanatory variables. The data has the form $(x_{1i}, x_{2i}, \cdots x_{ki})$ $i = 1, \cdots n$.

Accordingly the rv Y_i, $i = 1, \cdots n$ has the form

$$Y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \epsilon_i \quad (i = 1, \cdots n)$$

The errors are assumed to be independent and normally distributed such that

$$E[\epsilon_i] = 0, \quad Var(\epsilon_i) = \sigma^2 \quad i = 1, \cdots n$$

Therefore $E[Y_i] = \beta_0 + \beta x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki}$

$$\begin{aligned} V[Y_i] &= Var[Y \mid X = (x_1, \cdots x_k)] \\ &= Var[\beta_0 + \beta_1 x_{1i} + \cdots + \beta_k x_{ki} + \epsilon_i] \\ &= Var\, \epsilon_i = \sigma^2 \end{aligned} \quad (26)$$

RVs $Y_1, \cdots Y_k$ are mutually independent

Method of Least Squares

$$S(\beta_0, \cdots \beta_k) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{1i} \cdots - \beta_k x_{ki})^2$$

We get $(k+1)$ normal eqns to obtain least squares estimates $\hat{\beta}_0, \hat{\beta}_1, \cdots, \hat{\beta}_k$.

Example (Trivial)

We want to fit a 3-dimensional plane

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \qquad (27)$$

to a set of points $(y_i, x_{1i}, x_{2i}, x_{3i})$ $i=1,\dots n$

Empirical mean squared error is

$$S(\beta_0, \beta_1, \beta_2, \beta_3) = \sum_{i=1}^{n} \left[y_i - \beta_0 - \beta_1 x_{1i} - \beta_2 x_{2i} - \beta_3 x_{3i} \right]^2$$

To minimize S, we set partial derivatives w.r.t $\beta_0, \beta_1, \beta_2$ and β_3 equal to zero. This will yield 4-normal equations which will provide estimates of $\underline{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3)$.

Fitting polynomials of the form

$$y = a_0 + a_1 x + \dots + a_k x^k \qquad (28)$$
$$= \sum_{j=0}^{k} a_j x^j$$

<u>Note</u>: These polynomials are <u>linear</u> in unknown parameters $(a_0, a_1, \dots a_k)$ and hence gives linear system of equations for the unknowns.

- Nonlinear Regression — (Allen A.O)
- Regression with transformed Variables.

Note: We have considered polynomial regression

$$E[Y \mid X = x] = \beta_0 + \beta_1 x + \cdots + \beta_k x^k$$

This can be treated as part of linear regression

Regression with transformed variables.

(— this is of questionable value).

Ex 1 Original relationship $Y = \alpha e^{\beta X}$ (29)

Transformed $\longrightarrow Y' = (\ln \alpha) + \beta X$ (30)

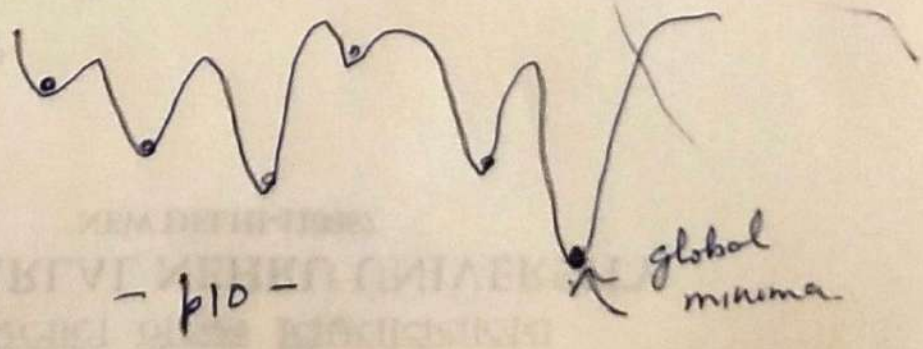
or $Y' = a + \beta X$, $Y' = \ln Y$

Linear in parameters

Ex 2 $y = a + b x^c$ (31)

logarithmic transformation will not help

$$G(a, b, c) = \sum_i (y_i - a - b x_i^c)^2 \quad (32)$$

To locate the 'global minimum'.

local minima